

SQL

In my database, I kept the non-countries. However, for this SQL query, I excluded it which can be seen in the SQL Queries I wrote to remove the non countries. I used this as a reference as to what countries to keep.

<https://www.state.gov/independent-states-in-the-world/>

Visualisation

This is my first attempt at Power BI so I am not entirely very familiar with the system nor did I want to overcomplicate it. I hope I did justice.

Q1

Explanation of Answer

SQL: I did a separate Excel Sheet extraction & calculation to check the dates for where I could get the data for as many countries as I could. 16/08/2021 had the least NULL values, thus the other two dates were picked. I hope I got it correct.

Visualisation: The values from the percentage change were from 0 to millions. Considering that it was regionally different and the large scale of difference between the values, it was hard to properly visualise it in a bar chart or until extra steps were taken to bin values or something similar.

Additionally, the million value was a major outlier. Thus two treemaps were used as part of this visualisation.

SQL

SELECT

'2021-03-01' AS Date_1,

loc.location AS Country_Name,

IFNULL(MAX(CASE WHEN g.date = '2021-03-01' THEN g.total_vaccinations END), 0) AS VOD1,

'2022-03-01' AS Date_2,

IFNULL(MAX(CASE WHEN g.date = '2022-03-01' THEN g.total_vaccinations END), 0) AS VOD2,

'2022-05-01' AS Date_3,

IFNULL(MAX(CASE WHEN g.date = '2022-05-01' THEN g.total_vaccinations END), 0) AS VOD3,

-- Calculate the Percentage Change of Totals with 0 for missing values

IFNULL(

CASE

WHEN MAX(CASE WHEN g.date = '2021-03-01' THEN g.total_vaccinations END) > 0

AND MAX(CASE WHEN g.date = '2022-03-01' THEN g.total_vaccinations END) > 0

```

THEN
(
(
(MAX(CASE WHEN g.date = '2022-03-01' THEN g.total_vaccinations END) -
MAX(CASE WHEN g.date = '2021-03-01' THEN g.total_vaccinations END)) /
NULLIF(MAX(CASE WHEN g.date = '2021-03-01' THEN g.total_vaccinations END), 0)
) -
(
(MAX(CASE WHEN g.date = '2022-05-01' THEN g.total_vaccinations END) -
MAX(CASE WHEN g.date = '2022-03-01' THEN g.total_vaccinations END)) /
NULLIF(MAX(CASE WHEN g.date = '2022-03-01' THEN g.total_vaccinations END), 0)
)
) * 100
ELSE 0
END,
0) AS Percentage_Change_of_Totals
FROM
A_Locations_Summary AS loc
LEFT JOIN
G_Vaccination_Population_Total_Data AS g
ON loc.loc_id = g.loc_id
AND g.date IN ('2021-03-01', '2022-03-01', '2022-05-01')
WHERE
loc.location NOT IN (
'Africa', 'Asia', 'Europe', 'European Union', 'Faeroe Islands', 'Falkland Islands',
'Gibraltar', 'Greenland', 'Hong Kong', 'Macao', 'New Caledonia', 'Northern Cyprus',
'Oceania', 'Palestine', 'Pitcairn', 'Scotland', 'Sint Maarten (Dutch part)', 'Tokelau',
'Turks and Caicos Islands', 'Wales', 'Wallis and Futuna', 'World', 'North America',
'South America', 'Aruba', 'Bermuda', 'Bonaire, Sint Eustatius, and Saba',
'British Virgin Islands', 'Cayman Islands', 'Cook Islands', 'Curacao', 'England',
'French Polynesia', 'Guernsey', 'Isle of Man', 'Jersey', 'Montserrat', 'Niue',

```

'Northern Ireland', 'Pitcairn Islands', 'Saint Helena', 'Taiwan', 'Timor'

)

GROUP BY

loc.location

ORDER BY

Percentage_Change_of_Totals DESC;

Snapshot

127.0.0.1:8080/query/

Final Assignment Test 3.db - Query

new_table_name Create Query

table name...

SELECT
'2021-03-01' AS Date_1,
loc.location AS Country_Name
IFNULL(MAX(CASE WHEN g.date = '2021-03-01' THEN g.total_vaccinations END), 0) AS VOD1,
'2022-03-01' AS Date_2,

Use Shift + Up/Down to navigate recently-executed queries

Execute Export JSON Export CSV SQL Help

Bookmarks

Results (190)

Permalink

Date_1	Country_Name	VOD1	Date_2	VOD2	Date_3	VOD3	Percentage_Change_of_Totals
2021-03-01	Thailand	319	2022-03-01	123999850	2022-05-01	133292376	38871300
2021-03-01	Japan	31785	2022-03-01	232946232	2022-05-01	274105909	732700
2021-03-01	South Korea	55673	2022-03-01	119365836	2022-05-01	123383182	214300
2021-03-01	Malaysia	34192	2022-03-01	67477866	2022-05-01	70490222	197200
2021-03-01	New Zealand	8132	2022-03-01	10651949	2022-05-01	11046234	130800
2021-03-01	Australia	41907	2022-03-01	54155508	2022-05-01	57735950	129100
2021-03-01	South Africa	73047	2022-03-01	31652555	2022-05-01	34877539	43200
2021-03-01	Montenegro	1531	2022-03-01	663653	2022-05-01	671916	43200
2021-03-01	Ecuador	77022	2022-03-01	31996832	2022-05-01	34496511	41400
2021-03-01	Uruguay	18483	2022-03-01	7659666	2022-05-01	8192574	41300
2021-03-01	Cambodia	87069	2022-03-01	35561224	2022-05-01	38390702	40700
2021-03-01	Zimbabwe	21713	2022-03-01	7901360	2022-05-01	10160810	36200
2021-03-01	Peru	332480	2022-03-01	63711347	2022-05-01	72914434	19000
2021-03-01	Albania	14295	2022-03-01	2708574	2022-05-01	2825500	18800
2021-03-01	India	14854136	2022-03-01	1776674379	2022-05-01	1890938196	11800
2021-03-01	Argentina	1349588	2022-03-01	94470969	2022-05-01	99417471	6800
2021-03-01	Brazil	8841027	2022-03-01	392925626	2022-05-01	429613287	4300
2021-03-01	Latvia	65733	2022-03-01	2854376	2022-05-01	2885344	4200
2021-03-01	Canada	1979597	2022-03-01	81283247	2022-05-01	84106133	4000
2021-03-01	Luxembourg	39134	2022-03-01	1270537	2022-05-01	1287546	3100
2021-03-01	Italy	4578311	2022-03-01	134289405	2022-05-01	136906764	2800

Visualisation



Q2

Explanation of Answer

SQL: I hope I have done justice to this calculation. Tried my best to get it right. Ordered the answer for best practice.

Visualisation: This is the most simplified approach. Alternatively, the data could be further processed by grouping countries, months or years and then visualising it in a similar manner. I didn't want to meddle with it further so I used this approach.

SQL

```
SELECT
```

```
loc.location AS "Country Name",
```

```
strftime('%m', a.date) AS "Month",
```

```
strftime('%Y', a.date) AS "Year",
```

```
-- Calculate the Growth Rate as a percentage and ensure it's not NULL, rounded to 2 decimal places
```

```
ROUND(IFNULL(((a.monthly_total - IFNULL(b.monthly_total, 0)) * 100.0 / IFNULL(b.monthly_total, 1)), 0), 2) AS "Growth rate of vaccine (GR)",
```

```
-- Calculate the Difference of Growth Rate to Global Average, rounded to 2 decimal places
```

```
ROUND(IFNULL(((a.monthly_total - IFNULL(b.monthly_total, 0)) * 100.0 / IFNULL(b.monthly_total, 1))
```

```
-
```

```
(SELECT AVG((c.monthly_total - IFNULL(d.monthly_total, 0)) * 100.0 / IFNULL(d.monthly_total, 1))
```

```
FROM (
```

```
SELECT loc_id,
```

```
strftime('%Y-%m', date) AS month_year,
```

```
SUM(daily_vaccinations) AS monthly_total
```

```
FROM F_Vaccination_Daily_Data
```

```
GROUP BY loc_id, strftime('%Y-%m', date)
```

```
) AS c
```

```
LEFT JOIN (
```

```
SELECT loc_id,
```

```

    strftime('%Y-%m', date) AS month_year,
    SUM(daily_vaccinations) AS monthly_total
FROM F_Vaccination_Daily_Data
GROUP BY loc_id, strftime('%Y-%m', date)
) AS d
ON c.loc_id = d.loc_id AND c.month_year = strftime('%Y-%m', DATE(d.month_year || '-01', '+1
month'))
WHERE c.month_year = strftime('%Y-%m', a.date)
), 0), 2) AS "Difference of growth rate to global average"

```

```

FROM (
SELECT loc_id,
strftime('%Y-%m', date) AS month_year,
SUM(daily_vaccinations) AS monthly_total,
date
FROM F_Vaccination_Daily_Data
GROUP BY loc_id, strftime('%Y-%m', date)
) AS a
LEFT JOIN (
SELECT loc_id,
strftime('%Y-%m', date) AS month_year,
SUM(daily_vaccinations) AS monthly_total
FROM F_Vaccination_Daily_Data
GROUP BY loc_id, strftime('%Y-%m', date)
) AS b
ON a.loc_id = b.loc_id AND a.month_year = strftime('%Y-%m', DATE(b.month_year || '-01', '+1
month'))

```

```

-- Join with A_Locations_Summary to get the actual country name
LEFT JOIN A_Locations_Summary AS loc
ON a.loc_id = loc.loc_id

```

-- Exclude the specified countries and filter for those above the global average

WHERE loc.location NOT IN (

'Anguilla', 'Africa', 'Asia', 'Europe', 'European Union', 'Faeroe Islands',

'Falkland Islands', 'Gibraltar', 'Greenland', 'Hong Kong', 'Macao',

'Northern Cyprus', 'Oceania', 'Palestine', 'Pitcairn', 'Scotland',

'Sint Maarten (Dutch part)', 'Tokelau', 'Turks and Caicos Islands',

'World', 'North America', 'South America', 'Aruba', 'Bermuda',

'Bonaire, Sint Eustatius, and Saba', 'British Virgin Islands',

'Cayman Islands', 'Cook Islands', 'Curacao', 'England',

'French Polynesia', 'Guernsey', 'Isle of Man', 'Jersey',

'Montserrat', 'New Caledonia', 'Niue', 'Northern Ireland',

'Pitcairn Islands', 'Saint Helena', 'Taiwan', 'Timor',

'Wales', 'Wallis and Futuna'

)

AND ROUND(IFNULL(((a.monthly_total - IFNULL(b.monthly_total, 0)) * 100.0 /
IFNULL(b.monthly_total, 1)), 0), 2) >

(SELECT AVG((c.monthly_total - IFNULL(d.monthly_total, 0)) * 100.0 / IFNULL(d.monthly_total, 1))

FROM (

SELECT loc_id,

strftime('%Y-%m', date) AS month_year,

SUM(daily_vaccinations) AS monthly_total

FROM F_Vaccination_Daily_Data

GROUP BY loc_id, strftime('%Y-%m', date)

) AS c

LEFT JOIN (

SELECT loc_id,

strftime('%Y-%m', date) AS month_year,

SUM(daily_vaccinations) AS monthly_total

FROM F_Vaccination_Daily_Data

GROUP BY loc_id, strftime('%Y-%m', date)

) AS d

ON c.loc_id = d.loc_id AND c.month_year = strftime('%Y-%m', DATE(d.month_year || '-01', '+1 month'))

WHERE c.month_year = strftime('%Y-%m', a.date)

)

ORDER BY "Country Name", "Year", "Month";

Snapshot

The screenshot shows a web-based SQL query interface. The browser address bar displays '127.0.0.1:8080/query/'. The page title is 'Final Assignment Test 3.db - Query'. On the left, there is a sidebar with a list of tables: A_Locations_Summary, B_Locations_Summary_D..., C_Vaccination_Manufac..., D_Vaccination_Age_Metrics, E_Vaccination_Summary, F_Vaccination_Daily_Data, G_Vaccination_Populat..., H_Vaccination_Populat..., I_CN_IN_IE_US_Summary, J_CN_IN_IE_US_Populat..., K_US_State_Summary, L_US_State_Vaccinatio..., M_US_State_Vaccinatio..., N_US_State_Vaccinatio..., and O_US_State_Vaccinatio... Below the list is a 'Toggle helper tables' link. The main query editor contains the following SQL code:

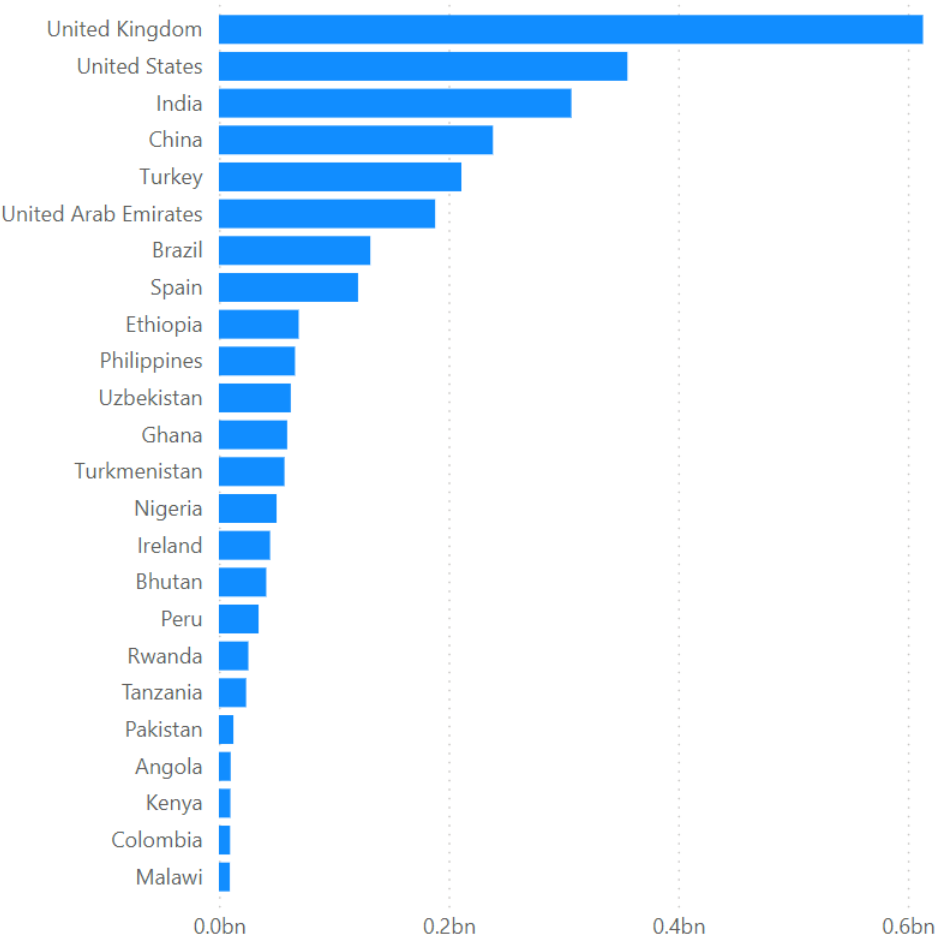
```
SELECT  
  loc.location AS "Country Name",  
  strftime('%m', a.date) AS "Month",  
  strftime('%Y', a.date) AS "Year",
```

 Below the query editor are buttons for 'Execute', 'Export JSON', 'Export CSV', and 'SQL Help'. The results section is titled 'Results (unable to determine)' and shows a table with the following columns: 'Country Name', 'Month', 'Year', 'Growth rate of vaccine (GR)', and 'Difference of growth rate to global average'. The table contains 20 rows of data for various countries including Afghanistan, Algeria, Andorra, Angola, Antigua and Barbuda, and Argentina. The bottom right corner of the interface features a 'Permalink' link and a small circular icon with the number '19'.

Country Name	Month	Year	Growth rate of vaccine (GR)	Difference of growth rate to global average
Afghanistan	07	2022	603.07	555.43
Afghanistan	08	2022	75.02	62.76
Afghanistan	02	2023	763.57	704.17
Afghanistan	05	2023	309.62	49.4
Afghanistan	06	2023	40.96	49.72
Afghanistan	08	2023	43.84	5.06
Algeria	04	2022	723.33	713.02
Andorra	12	2021	691.12	589.33
Andorra	01	2022	65.98	21.9
Angola	03	2021	12361900.0	10225323.51
Angola	09	2021	324.8	171.24
Angola	02	2023	87.24	27.85
Angola	06	2023	18.84	27.61
Angola	07	2023	9.2	9.95
Angola	12	2023	-15.69	12.83
Antigua and Barbuda	02	2021	1074000.0	92489.94
Antigua and Barbuda	09	2021	248.61	95.04
Antigua and Barbuda	06	2022	340.38	125.59
Antigua and Barbuda	07	2022	94.03	46.4
Antigua and Barbuda	08	2022	260.69	248.43
Argentina	12	2022	303.9	144.85

Visualisation

Country Name



Sum of Difference of growth rate to global average

Q3

Explanation of Answer

SQL: In my initial calculation where I excluded the top 5, the total percentage value per country was 100% but when I ensured that it was for the top 5, the total percentage per country was at 100%. I hope it's in line with the Q cause it aided in the Visualisations.

Visualisation: For each country, all the values added up to 100%. Thus stacked bar chart felt useful. It could be used vertically instead of horizontally too. Tried to be as thorough with the data presentation as possible.

SQL

SELECT

vt."Vaccine Type",

loc.location AS Country,

ROUND((vt.total_vaccine_vaccinations * 100.0 / ct.total_country_vaccinations), 2) AS "Percentage of Vaccine Type"

FROM

(SELECT

loc_id AS Country,

vaccine AS "Vaccine Type",

SUM(total_vaccinations) AS total_vaccine_vaccinations

FROM

C_Vaccination_Manufacturers

GROUP BY

loc_id, vaccine

) AS vt

JOIN

(SELECT

loc_id AS Country,

SUM(total_vaccinations) AS total_country_vaccinations

FROM

C_Vaccination_Manufacturers

```

GROUP BY

loc_id

) AS ct

ON vt.Country = ct.Country

JOIN

A_Locations_Summary loc

ON vt.Country = loc.loc_id

WHERE

loc.location NOT IN (

'Africa', 'Asia', 'Europe', 'European Union', 'Faeroe Islands',

'Falkland Islands', 'Gibraltar', 'Greenland', 'Hong Kong', 'Macao',

'Northern Cyprus', 'Oceania', 'Palestine', 'Pitcairn', 'Scotland',

'Sint Maarten (Dutch part)', 'Tokelau', 'Turks and Caicos Islands',

'World', 'North America', 'South America', 'Aruba', 'Bermuda',

'Bonaire, Sint Eustatius, and Saba', 'British Virgin Islands',

'Cayman Islands', 'Cook Islands', 'Curacao', 'England',

'French Polynesia', 'Guernsey', 'Isle of Man', 'Jersey',

'Montserrat', 'New Caledonia', 'Niue', 'Northern Ireland',

'Pitcairn Islands', 'Saint Helena', 'Taiwan', 'Timor',

'Wales', 'Wallis and Futuna'

)

AND (

SELECT COUNT(*)

FROM (

SELECT

loc_id AS Country,

vaccine AS "Vaccine Type",

SUM(total_vaccinations) AS total_vaccine_vaccinations

FROM

C_Vaccination_Manufacturers

GROUP BY

```

```

loc_id, vaccine
) AS sub_vp

JOIN

(SELECT

loc_id AS Country,

SUM(total_vaccinations) AS total_country_vaccinations

FROM

C_Vaccination_Manufacturers

GROUP BY

loc_id

) AS sub_ct

ON sub_vp.Country = sub_ct.Country

WHERE sub_vp.Country = vt.Country

AND (sub_vp.total_vaccine_vaccinations * 100.0 / sub_ct.total_country_vaccinations) >=

(vt.total_vaccine_vaccinations * 100.0 / ct.total_country_vaccinations)

) <= 5

ORDER BY

loc.location, "Percentage of Vaccine Type" DESC;

```

Snapshot

The screenshot shows a web-based SQL query editor interface. The browser address bar displays '127.0.0.1:8080/query/'. The page title is 'Final Assignment Test 3.db - Query'. On the left, there is a sidebar with a list of tables: A_Locations_Summary, B_Locations_Summary_D..., C_Vaccination_Manufac..., D_Vaccination_Age_Metrics, E_Vaccination_Summary, F_Vaccination_Daily_Data, G_Vaccination_Populat..., H_Vaccination_Populat..., I_CN_IN_IE_US_Summary, J_CN_IN_IE_US_Populat..., K_US_State_Summary, L_US_State_Vaccinatio..., M_US_State_Vaccinatio..., N_US_State_Vaccinatio..., and O_US_State_Vaccinatio... Below this list is a 'Toggle helper tables' link. The main area contains a SQL query in a text editor. The query is a complex join and aggregation query. Below the query editor, there are buttons for 'Execute', 'Export JSON', 'Export CSV', and 'SQL Help'. To the right of the query editor, there is a 'new_table_name' input field, a 'Create' button, and a 'Query' button. Below the query editor, there is a 'Results (unable to determine)' section. This section contains a table with three columns: 'Vaccine Type', 'Country', and 'Percentage of Vaccine Type'. The table lists various vaccine types and their corresponding countries and percentages. The results are sorted by 'Percentage of Vaccine Type' in descending order. The table has 20 rows of data. The bottom right corner of the interface shows a 'Permalink' link and a small icon.

table name...

SELECT

vt "Vaccine Type",

loc.location AS Country,

ROUND(vt.total_vaccine_vaccinations * 100.0 / ct.total_country_vaccinations, 2) AS "Percentage of Vaccine Type"

FROM

Use Shift + Up/Down to navigate recently-executed queries

Execute Export JSON Export CSV SQL Help

Bookmarks

Results (unable to determine)

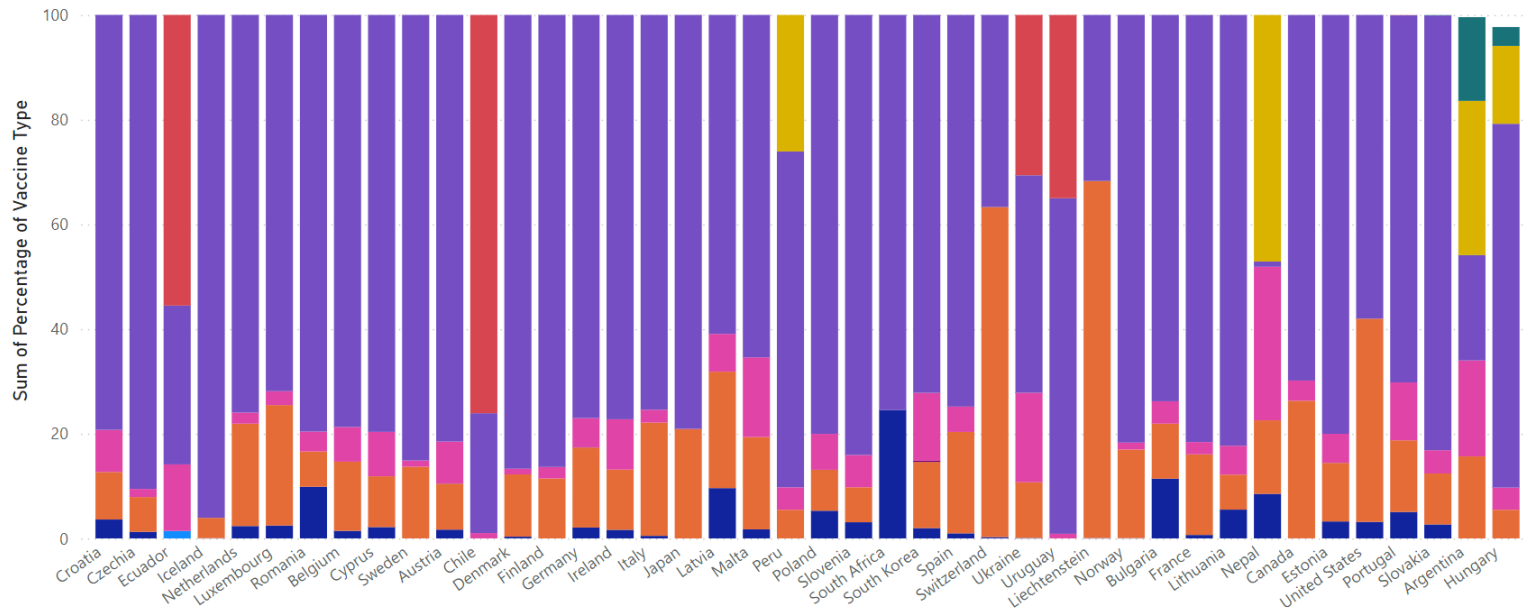
Permalink

Vaccine Type	Country	Percentage of Vaccine Type
Sinopharm/Beijing	Argentina	29.46
Pfizer/BioNTech	Argentina	20.12
Oxford/AstraZeneca	Argentina	18.33
Sputnik V	Argentina	16.0
Moderna	Argentina	15.67
Pfizer/BioNTech	Austria	81.5
Moderna	Austria	8.77
Oxford/AstraZeneca	Austria	8.0
Johnson&Johnson	Austria	1.68
Novavax	Austria	0.05
Pfizer/BioNTech	Belgium	78.72
Moderna	Belgium	13.21
Oxford/AstraZeneca	Belgium	6.62
Johnson&Johnson	Belgium	1.45
Novavax	Belgium	0.0
Pfizer/BioNTech	Bulgaria	73.79
Johnson&Johnson	Bulgaria	11.41
Moderna	Bulgaria	10.51
Oxford/AstraZeneca	Bulgaria	4.28
Pfizer/BioNTech	Canada	69.83
Moderna	Canada	26.29
Oxford/AstraZeneca	Canada	3.85

Visualisation

< Back to report | SUM OF PERCENTAGE OF VACCINE TYPE BY COUNTRY AND VACCINE TYPE

Vaccine Type CanSino Johnson&Johnson Moderna Novavax Oxford/AstraZeneca Pfizer/BioNTech Sinopharm/Beijing Sinovac Sputnik V



Q4

Explanation of Answer

SQL: To the marker, the discussion forum was flooded with this Q and frankly, while I attempted but failed at a 2nd Query. My logic was total vaccines per country based on data source & month. I also used data from B_Locations_Summary_Details table while excluding I_CN, IN, IE, US_Summary to avoid duplication & data inconsistency as the data from China, India, Ireland & USA were also present in the Locations table.

Visualisation: All countries were present in this data. There are however limitations to data visualised thus the top 25 countries were visualised.

SQL

SELECT

loc.location AS "Country Name",

strftime('%Y-%m', g.date) AS "Month",

details.source_name || ' (' || details.source_url || ')' AS "Source Name (URL)",

IFNULL(SUM(g.total_vaccinations), 0) AS "Total Administered Vaccines"

FROM

G_Vaccination_Population_Total_Data AS g

JOIN

A_Locations_Summary AS loc ON g.loc_id = loc.loc_id

JOIN

B_Locations_Summary_Details AS details ON g.loc_id = details.loc_id

AND strftime('%Y-%m', g.date) = strftime('%Y-%m', details.last_observation_date)

WHERE

loc.location NOT IN (

'Africa', 'Asia', 'Europe', 'European Union', 'Faeroe Islands',

'Falkland Islands', 'Gibraltar', 'Greenland', 'Hong Kong', 'Macao',

'Northern Cyprus', 'Oceania', 'Palestine', 'Pitcairn', 'Scotland',

'Sint Maarten (Dutch part)', 'Tokelau', 'Turks and Caicos Islands',

'World', 'North America', 'South America', 'Aruba', 'Bermuda',

'Bonaire, Sint Eustatius, and Saba', 'British Virgin Islands',

'Cayman Islands', 'Cook Islands', 'Curacao', 'England',

'French Polynesia', 'Guernsey', 'Isle of Man', 'Jersey',
'Montserrat', 'New Caledonia', 'Niue', 'Northern Ireland',
'Pitcairn Islands', 'Saint Helena', 'Taiwan', 'Timor',
'Wales', 'Wallis and Futuna'
)

GROUP BY

loc.location,

strftime('%Y-%m', g.date),

details.source_name,

details.source_url

ORDER BY

"Country Name" ASC,

"Month" DESC;

Snapshot

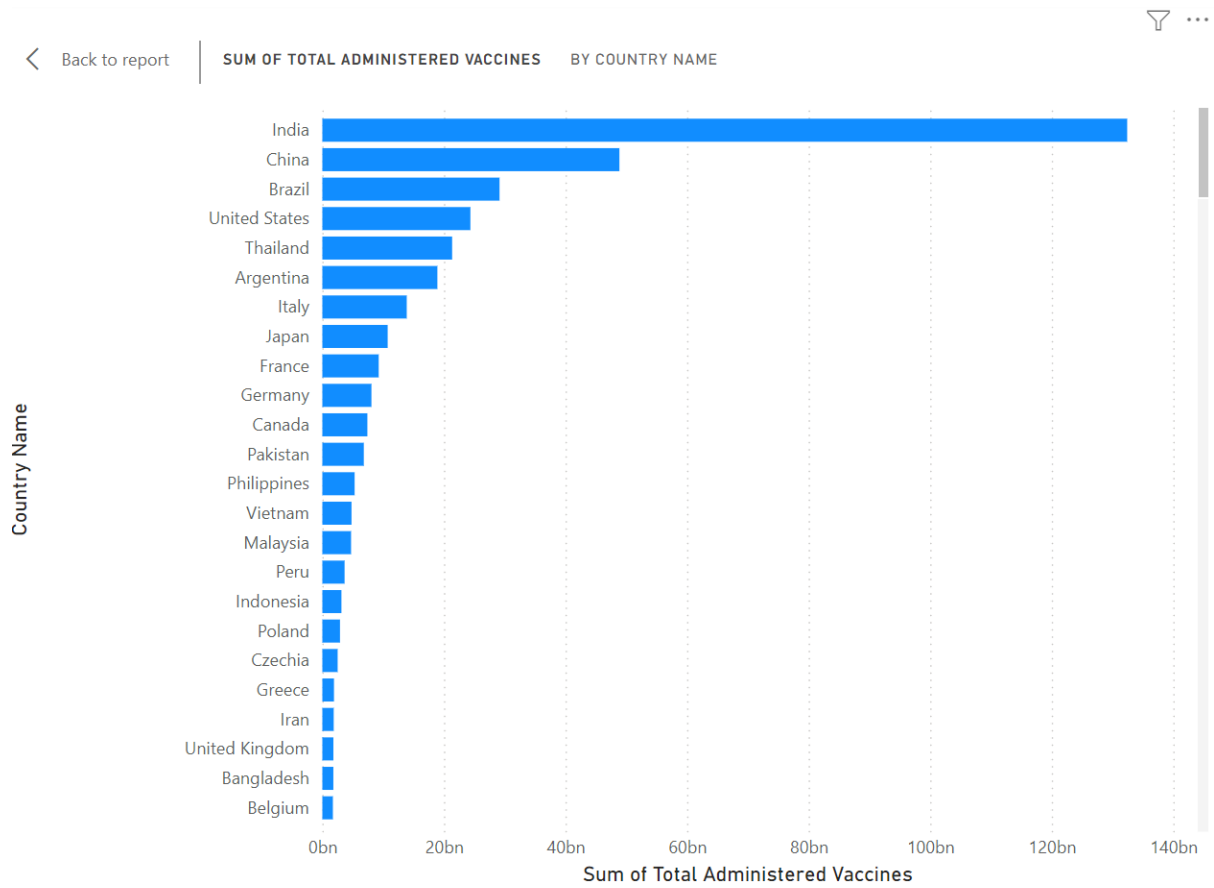
The screenshot shows a web application interface for a database query. The browser address bar displays '127.0.0.1:8080/query/'. The page title is 'Final Assignment Test 3.db - Query'. On the left, there is a sidebar with a list of tables: 'A_Locations_Summary', 'B_Locations_Summary_D...', 'C_Vaccination_Manufac...', 'D_Vaccination_Age_Metrics', 'E_Vaccination_Summary', 'F_Vaccination_Daily_Data', 'G_Vaccination_Populat...', 'H_Vaccination_Populat...', 'I_CN_IN_IE_US_Summary', 'J_CN_IN_IE_US_Populat...', 'K_US_State_Summary', 'L_US_State_Vaccinatio...', 'M_US_State_Vaccinatio...', 'N_US_State_Vaccinatio...', and 'O_US_State_Vaccinatio...'. Below the table list is a 'Toggle helper tables' link. The main area contains a SQL query editor with the following query:

```
SELECT  
  loc.location AS "Country Name",  
  strftime('%Y-%m', g.date) AS "Month",  
  details.source_name || ' (' || details.source_url || ')' AS "Source Name (URL)",  
  IFNULL(SUM(g.total_vaccinations), 0) AS "Total Administered Vaccines"
```

 Below the query editor are buttons for 'Execute', 'Export JSON', 'Export CSV', and 'SQL Help'. To the right of the query editor is a 'Bookmarks' dropdown. Below the query editor, the results are displayed under the heading 'Results (unable to determine)'. The results are shown in a table with the following columns: 'Country Name', 'Month', 'Source Name (URL)', and 'Total Administered Vaccines'. The table contains 20 rows of data, including countries like Afghanistan, Albania, Algeria, Andorra, Angola, Anguilla, Antigua and Barbuda, Argentina, Armenia, Australia, Austria, Azerbaijan, Bahamas, Bahrain, Bangladesh, Barbados, Belarus, Belgium, Belize, Benin, and Bhutan. The 'Total Administered Vaccines' column shows values ranging from 0 to 229647500.

Country Name	Month	Source Name (URL)	Total Administered Vaccines
Afghanistan	2023-12	World Health Organization (https://data.who.int/da ...)	229647500
Albania	2023-09	World Health Organization (https://data.who.int/da ...)	12355864
Algeria	2022-09	World Health Organization (https://data.who.int/da ...)	61069768
Andorra	2023-09	World Health Organization (https://data.who.int/da ...)	471216
Angola	2023-12	World Health Organization (https://data.who.int/da ...)	27819132
Anguilla	2023-03	World Health Organization (https://ais.paho.org/im ...)	98416
Antigua and Barbuda	2022-09	Ministry of Health (https://covid19.who.int/)	682560
Argentina	2024-01	Ministry of Health (https://covidstats.com.ar/)	18922840668
Armenia	2023-10	World Health Organization (https://data.who.int/da ...)	20312271
Australia	2023-12	Government of Australia via CovidBaseAU (https://d ...)	694965485
Austria	2023-06	European CDC (https://www.ecdc.europa.eu/en/public ...)	716373945
Azerbaijan	2022-08	Government of Azerbaijan (https://koronavirusinfo. ...)	940721724
Bahamas	2023-05	Pan American Health Organization (https://ais.paho ...)	1099077
Bahrain	2022-12	Ministry of Health (https://data.who.int/dashboard ...)	69526570
Bangladesh	2024-07	Directorate General of Health Services (http://103 ...)	1808712770
Barbados	2023-12	Ministry of Health (https://data.who.int/dashboard ...)	1536196
Belarus	2023-09	World Health Organization (https://data.who.int/da ...)	90044000
Belgium	2024-02	Sciensano (https://epistat.wiv-isp.be/covid/)	1730796210
Belize	2023-05	World Health Organization (https://ais.paho.org/im ...)	0
Benin	2022-12	Ministry of Health (https://data.who.int/dashboard ...)	16930164
Bhutan	2022-10	World Health Organization (https://data.who.int/da ...)	40177432

Visualisation



Q5

Explanation of Answer

SQL: I am not sure if AS "United States" may be the logic used but it is to ensure that the data matches the naming convention required.

Visualisation: I switched to the normal PIVOT Table on Excel due to my familiarity and I did a time series using Quarterly breaks (easier to capture 2 years info) and summing the incremental/decremental values to give an idea of the speed taken over the period. The figures in the data present has a large difference from being in the hundreds range to the millions range.

SQL

SELECT

a.date AS "Dates",

-- Increment for United States

(us.people_fully_vaccinated - COALESCE(prev_us.people_fully_vaccinated, 0)) AS "United States",

-- Increment for China

(cn.people_fully_vaccinated - COALESCE(prev_cn.people_fully_vaccinated, 0)) AS "China",

-- Increment for Ireland

(ie.people_fully_vaccinated - COALESCE(prev_ie.people_fully_vaccinated, 0)) AS "Ireland",

-- Increment for India

(inr.people_fully_vaccinated - COALESCE(prev_inr.people_fully_vaccinated, 0)) AS "India"

FROM

-- Selecting unique dates in the specified period

(SELECT DISTINCT date FROM J_CN_IN_IE_US_Population_Total

WHERE date BETWEEN '2022-01-01' AND '2023-12-31') AS a

-- United States current and previous day join

LEFT JOIN J_CN_IN_IE_US_Population_Total AS us

ON a.date = us.date AND us.loc_id = 'USA'

LEFT JOIN J_CN_IN_IE_US_Population_Total AS prev_us

ON us.loc_id = prev_us.loc_id

AND prev_us.date = (SELECT MAX(date) FROM J_CN_IN_IE_US_Population_Total

WHERE loc_id = 'USA' AND date < us.date)

-- China current and previous day join

LEFT JOIN J_CN_IN_IE_US_Population_Total AS cn

ON a.date = cn.date AND cn.loc_id = 'CHN'

LEFT JOIN J_CN_IN_IE_US_Population_Total AS prev_cn

ON cn.loc_id = prev_cn.loc_id

AND prev_cn.date = (SELECT MAX(date) FROM J_CN_IN_IE_US_Population_Total

WHERE loc_id = 'CHN' AND date < cn.date)

-- Ireland current and previous day join

LEFT JOIN J_CN_IN_IE_US_Population_Total AS ie

ON a.date = ie.date AND ie.loc_id = 'IRL'

LEFT JOIN J_CN_IN_IE_US_Population_Total AS prev_ie

ON ie.loc_id = prev_ie.loc_id

AND prev_ie.date = (SELECT MAX(date) FROM J_CN_IN_IE_US_Population_Total

WHERE loc_id = 'IRL' AND date < ie.date)

-- India current and previous day join

LEFT JOIN J_CN_IN_IE_US_Population_Total AS inr

ON a.date = inr.date AND inr.loc_id = 'IND'

LEFT JOIN J_CN_IN_IE_US_Population_Total AS prev_inr

ON inr.loc_id = prev_inr.loc_id

AND prev_inr.date = (SELECT MAX(date) FROM J_CN_IN_IE_US_Population_Total

WHERE loc_id = 'IND' AND date < inr.date)

ORDER BY a.date;

Snapshot

127.0.0.1:8080/query/

Final Assignment Test 3.db - Query

new_table_name [Create](#) [Query](#)

table name...

[A_Locations_Summary](#)
[B_Locations_Summary_D...](#)
[C_Vaccination_Manufac...](#)
[D_Vaccination_Age_Metrics](#)
[E_Vaccination_Summary](#)
[F_Vaccination_Daily_Data](#)
[G_Vaccination_Populat...](#)
[H_Vaccination_Populat...](#)
[I_CN_IN_IE_US_Summary](#)
[J_CN_IN_IE_US_Populat...](#)
[K_US_State_Summary](#)
[L_US_State_Vaccinatio...](#)
[M_US_State_Vaccinatio...](#)
[N_US_State_Vaccinatio...](#)
[O_US_State_Vaccinatio...](#)

[Toggle helper tables](#)

```

SELECT
  a.date AS "Dates",
  -- Increment for United States
  (us.people_fully_vaccinated - COALESCE(prev_us.people_fully_vaccinated, 0)) AS "United States",
  (cn.people_fully_vaccinated - COALESCE(prev_cn.people_fully_vaccinated, 0)) AS "China",
  (ie.people_fully_vaccinated - COALESCE(prev_ie.people_fully_vaccinated, 0)) AS "Ireland",
  (in.people_fully_vaccinated - COALESCE(prev_in.people_fully_vaccinated, 0)) AS "India"
FROM A_Locations_Summary a
LEFT JOIN I_CN_IN_IE_US_Summary i ON a.location = i.location
LEFT JOIN L_US_State_Vaccination l ON a.location = l.location
LEFT JOIN M_US_State_Vaccination m ON a.location = m.location
LEFT JOIN N_US_State_Vaccination n ON a.location = n.location
LEFT JOIN O_US_State_Vaccination o ON a.location = o.location

```

Use Shift + Up/Down to navigate recently-executed queries

[Execute](#) [Export JSON](#) [Export CSV](#) [SQL Help](#)

Bookmarks

Results (unable to determine) [Permalink](#)

Dates	United States	China	Ireland	India
2022-01-01	17847	0	106	3869210
2022-01-02	91704	0	631	1826897
2022-01-03	200389	0	733	3034271
2022-01-04	216928	0	796	2585550
2022-01-05	216178	0	920	3195789
2022-01-06	216124	1210000000	1703	3164770
2022-01-07	250179	5878000	790	5064540
2022-01-08	177888	-1215878000	750	1109853
2022-01-09	76478	0	667	4424499
2022-01-10	193612	0	517	2764059
2022-01-11	204184	0	1377	5852587
2022-01-12	204066	0	933	2952709
2022-01-13	191844	0	820	4516193
2022-01-14	177308	1220584000	844	1291354
2022-01-15	88356	-1220584000	659	4635942
2022-01-16	58529	0	1218	2497849
2022-01-17	141931	0	565	2602607
2022-01-18	191442	0	461	151381
2022-01-19	189588	0	628	6786162
2022-01-20	179487	0	943	4301942
2022-01-21	164601	1234450000	978	5187087

Visualisation

